

2.4 Introduction to Expert System

2.4.1 Definition of Expert System

Following are various definitions of expert systems :-

Definition 1 - Expert system is a artificial intelligence based system that converts the knowledge of an expert in a specific subject into a software code. This code can be merged with other such codes based on the knowledge of other experts, and used for answering queries submitted through a computer.

Definition 2 - Expert system is a piece of software which uses databases of expert knowledge to offer advice or make decisions in such areas as medical diagnosis.

Definition 3 - Expert system is a computer program that contains a knowledgebase and a set of algorithms or rules that infer new facts from knowledge and from incoming data. An expert system is an AI application that uses a knowledgebase of human expertise to aid in solving problems.

Definition 4 - Expert system is a model and associated procedure that exhibits, within a specific domain, a degree of expertise in problem solving that is comparable to a human expert.

Definition 5 - Expert system is a computing system capable of representing and reasoning about some knowledge rich domain, which usually requires a human expert, with a view towards solving problems and/or giving advice. Its level of performance makes it "expert".

Definition 6 - Expert system = Knowledge + Inference engine.

All in all, an expert system contains knowledge acquired by interviewing human expertise in some domain. Expert system can not operate in situations that call for common sense. Thus, Expert system perform a task in limited domains that require human expertise like medical diagnosis, fault diagnosis, status monitoring, data interpretation, mineral exploration, tutoring, computer configuration, credit checking etc.

2.4.2 Common Sense Reasoning

Computers have an entirely deserved reputation for lacking common - sense.

A human expert has two main types of knowledge -

- i) Domain - specific knowledge
- ii) Common - sense knowledge

Domain - specific knowledge deals with all kinds of knowledge about a particular domain. All other pieces of knowledge that help in reasoning other than domain-specific knowledge are termed as common-sense knowledge.

Common - sense knowledge when coupled, with domain - specific knowledge increases the performance capability of the system.

There are number of techniques available that can be used to enable an AI program to represent and reason with common - sense knowledge.

As of yet, however, no program can match the common-sense reasoning powers of a five-years-old-child.

To understand the common - sense reasoning consider the following real life situation.

Ram enjoys shopping in the mall only when the stores are not crowded. He has agreed to accompany Shyam there on the following Wednesday evening since this is normally a time when few people at shop.

Before the given data, several of the larger stores announce a one - time, special sale starting on that Wednesday evening.

Ram fearing large crowds, now retracts the offer to accompany Shyam, promising to go on some future data.

On the Tuesday before the sale was to commence, whether forecasts predicted heavy snow.

Now believing the whether would discourage most shoppers, Ram once again agreed to join Shyam. But, unexpectedly, on the given Wednesday, the forecasts proved to be false; so Ram once again declined to go.

This anecdote illustrates that how one's beliefs can change in a dynamic environment.

And while one's beliefs may not fluctuate as much as Ram's in most situations.

This form of belief is not uncommon. Indeed it is common enough that we label it as a form of common-sense reasoning i.e. reasoning with uncertain knowledge.

2.4.3 Why to Use an Expert System ?

Expert System (ES) helps to preserve knowledge. ES can be well suitable and affordable when expertise is expensive or unavailable. Human can fail to work under pressure and time bound but system would not. ES can train new employees thereby saving expertise time.

ES can save cost, as it can perform same task repeatedly and thereby providing good quality and uniform performance. ES can provide rigid solution for large problems.

2.4.4 Is Expert System Feasible ?

The major factor to consider for feasibility is **the cost**. How much is the need of ES would justify the cost to be incurred ?

When human expertise is not available ES is best solution in all criterion and conditions.

The major problem faced by ES is what is the size and scope of the system. This is relative to resources and evolving technology.

ES implementation is structured and does not require any common sense reasoning and hence perform uniformly with respect to all situations.

2.4.5 Advantages and Disadvantages of Expert System

Advantages of expert system

1. A well functioning ES provides increased distribution of expertise.
2. ES gives broader job description for individual workers.
3. ES can work round the clock.
4. ES can be used by the user more frequently.
5. ES can be trained faster as compare to human, therefore training becomes less time and money consuming.
6. ES enables new services.
7. ES work as a new communication channel for knowledge.
8. ES holds and maintains significant levels of information.
9. ES has faster adaptation to changing conditions.
10. ES provides consistent answers for repetitive decisions, processes and tasks.
11. ES encourages organizations to clarify the logic of their decision-making.
12. ES never "forgets" to ask a question, as a human might.
13. A multi-user expert system can serve more users at a time.

Disadvantages of expert system

1. ES lacks common sense needed in some decision making.
2. ES can not make creative responses as human expert would in unusual circumstances.
3. Domain experts may not be always able to explain their logic and reasoning.
4. When the problem is outside the domain of ES then ES can not respond reasonably well when no answer exists.
5. Errors may occur in the knowledgebase, and lead to wrong decisions by ES.

6. ES can not adapt to changing environments, unless knowledgebase is changed. That is, ES can not be made to do self learning.
7. ES is dependent on symbolic input only and can not respond to sensory experience as human can do.

2.4.6 Limitations of ES

- Knowledge is not always readily available.
- It can be difficult to extract expertise from humans.
- There are frequently multiple correct assessments.
- Time deadline can be a pressure on ES while processing.
- Users have cognitive limits.
- ES works well only within a narrow domain of knowledge.
- Most experts do not have an independent means to validate results.
- Vocabulary of knowledge is often limited and difficult to understand.
- Help from knowledge engineers is difficult to obtain and costly too.
- Potential for lack of trust on the part of the end-users.
- Knowledge transfer is subject to biases.

2.4.7 The Success Factors of ES

- Level of knowledge must be sufficiently high.
- Expertise must be available from at least one expert.
- The problem to be solved must be fuzzy.
- The problem must be narrow in scope.
- The shell must be of high quality and naturally store and manipulate the knowledge.
- The users interface must be friendly to novice users.
- The problem to be solved must be difficult and important enough to justify the development of a system.
- Knowledgeable developers with good people skills are needed.
- The impact of the ES must be considered.
- The impact should be favorable.
- Management support is needed.

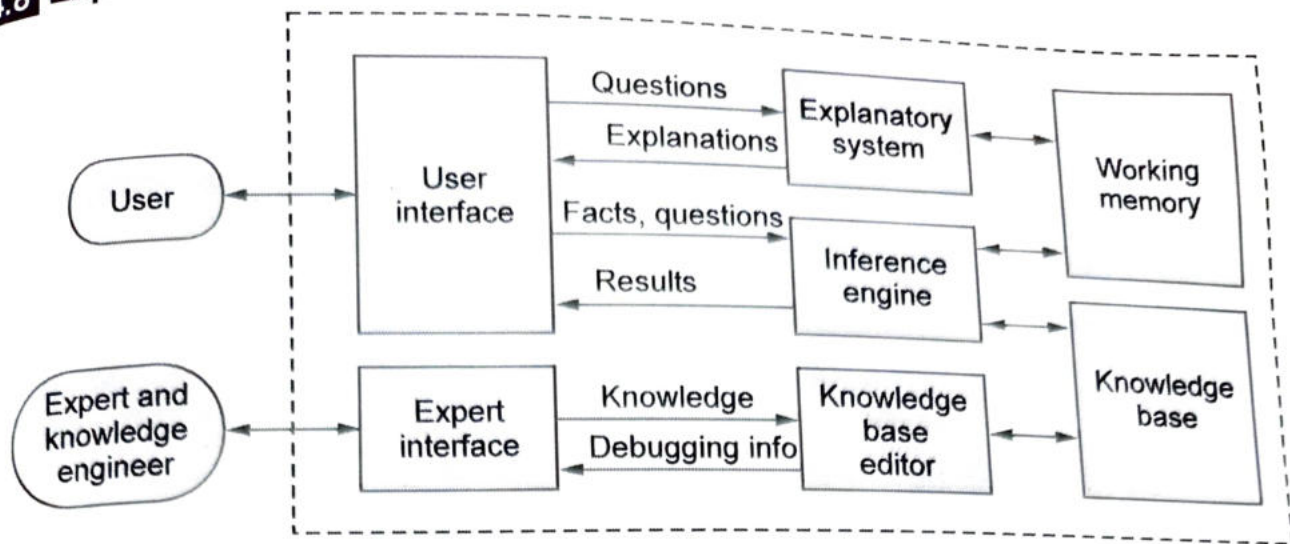


Fig. 2.4.1 Constitute of domain - specific and general knowledge section

The user interacts with the system through a *user interface* which may use menus, natural language or any other style of interaction. Then an *inference engine* is used to reason with both the *expert knowledge* (extracted from the expert) and data specific to the particular problem being solved. The expert knowledge will typically be in the form of a set of IF-THEN rules. The *case specific data* includes both data provided by the user and partial conclusions (along with certainty measures) based on this data. In a simple forward chaining rule-based system the case specific data will be the elements in *working memory*.

Every expert system consists of two principle parts : The knowledge base; and the reasoning or inference engine. The *knowledge base* of expert systems contains both factual and heuristic knowledge. *Factual knowledge* is that knowledge of the task domain that is widely shared, typically found in textbooks or journals, and commonly agreed upon by those knowledgeable in the particular field. *Heuristic knowledge* is the less rigorous, gained from experience, more judgemental knowledge of performance. In contrast to factual knowledge, heuristic knowledge is rarely discussed, and is largely individualistic. It is the knowledge of good practice, good judgement, and plausible reasoning in the field. It is the knowledge that underlies the "art of good guessing".

Knowledge representation formalizes and organizes the knowledge. One widely used representation is the production rule, or simply rule. A rule consists of an IF part and a THEN part (also called a *condition* and an *action*). The IF part lists a set of conditions in some logical combination. The piece of knowledge represented by the production rule is relevant to the line of reasoning being developed if the IF part of the rule is satisfied; consequently, the THEN part can be concluded, or its problem-solving action taken. Expert systems whose knowledge is represented in rule form are called *rule-based systems*.

Generally the expert systems also have an *explanation subsystem*, which allows the program to explain its reasoning to the user. Some systems also have a *knowledge base editor* which help the expert or knowledge engineer to easily update and check the knowledge base.

Another important feature of expert systems is the way they (usually) separate domain specific knowledge from more general purpose reasoning and representation techniques. The general purpose module (in the dotted box in the figure) is referred to as an *expert system shell*. As seen in the Fig. 2.4.1, the shell will provide the inference engine (and knowledge representation scheme), a user interface, an explanation system and sometimes a knowledge base editor. Given a new kind of problem to solve (say, TV set design), one can usually find a shell that provides the right sort of support for that problem, so what is needed to do is provide the expert knowledge. There are numerous commercial expert system shells, each one appropriate for a slightly different range of problems. (Expert systems work in industry includes both writing expert system shells and writing expert systems using shells). Using shells to write expert systems generally greatly reduces the cost and time development (compared with writing the expert system from scratch).

An ideal expert system should include the following :-

- Extensive specific knowledge from the domain of interest.
- Extensive database interfaces.
- Application of several techniques.
- Support for heuristic analysis.
- Capacity to infer new knowledge from existing knowledge.
- Decisions under uncertainty.
- Decisions with unknowns.
- Symbolic processing.
- An ability to explain its own reasoning.

Types of architecture

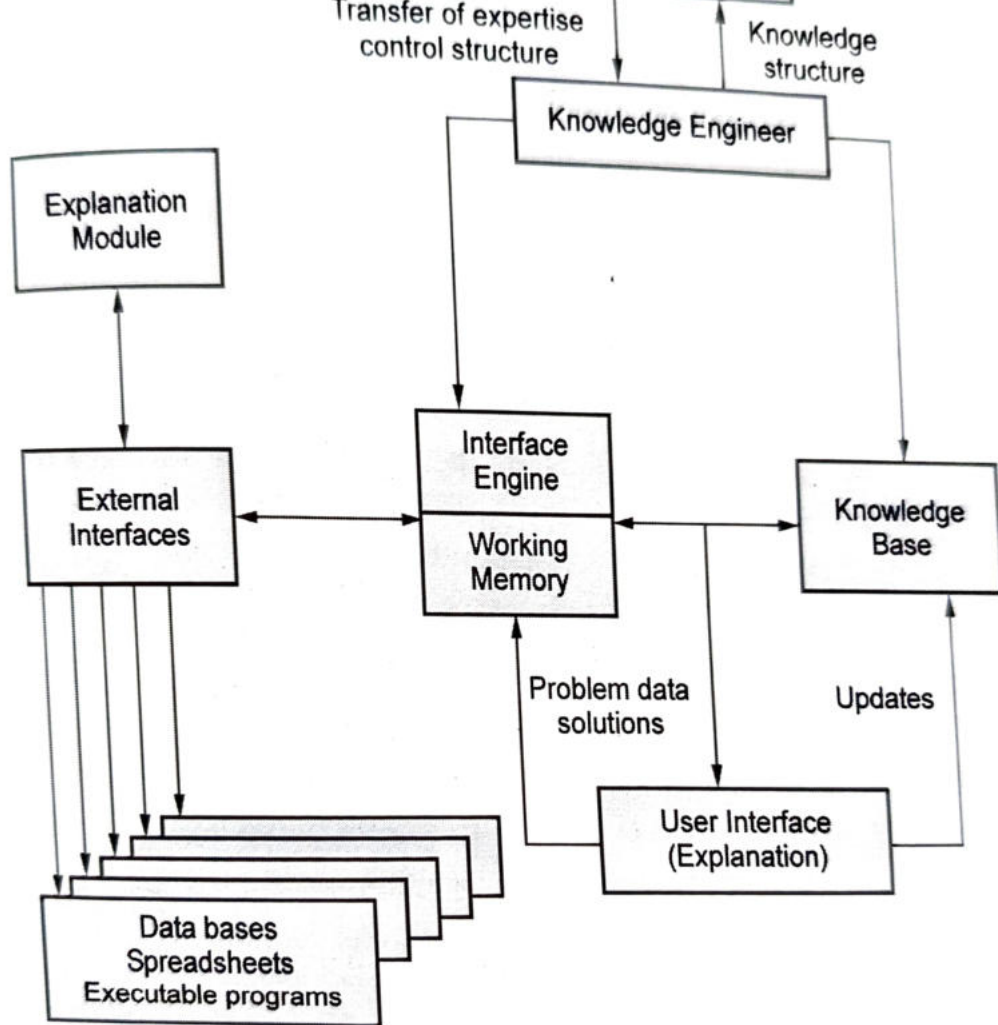
i) Rule Based Architecture of an Expert System

The most common form of architecture used in expert and other types of knowledge based systems is the production system or it is called rule based systems. This type of system uses knowledge encoded in the form of production rules i.e. if-then rules. The rule has a conditional part on the left hand side and a conclusion or action part on the right hand side. For example
if: condition1 and condition2 and condition3

Then : Take action4



Domain Expert



systems components

The components of the rule based archi

1. **User interface** : It is the mechanism by which the

communicate with each other i.e. the user interacts with the system through a user interface. It acts as a bridge between user and expert system. This module accepts the user queries and submits those to the expert system. The user normally consults the expert system for following reasons.

- a) To get answer of his/her queries.
- b) To get explanation about the solution for psychological satisfaction.

The user interface module is designed in such a way that at user level it accepts the query in a language understandable by expert system. To make the expert system user friendly, the user interface interacts with the user in natural language. The user interface provides as much facilities as possible such as menus, graphical interfaces etc. to make the dialog user friendly and more attractive.

2. Explanation module : The explanation module explains the reasoning of the system to a user. It provides the user with an explanation of the reasoning process when requested. The credibility of expert system will be established only when it is able to explain "how and why" a particular conclusion is drawn. This explanation increases the belief of user in the expert system.

a) Explanation (How) : To respond to a how query, the explanation module traces the chain of rules fired during a consultation with the user. This explanation mode can be activated once the process is over. It explains how a fact was deduced by the system and similarly how a rule was/wasn't used. The simplest way to specify this is to explain the rule which allows the deduction. For e.g.

If the system (S) will give information about the parent-child relationship to the user (U) then the followings can be possible.

S: My diagnosis is "A is the father of B"

U: How?

S: The result was obtained by the addition of following facts and rules.

Fact no 11: A is the parent of Hari.

Fact no 15: A is a male.

Fact no 110: X is father of Y;

X is parent of Y, X is male.

So A is the father of B.

b) Explanation (Why) ? To respond to a why query, the explanation module must be able to explain why certain information is needed by the inference engine to complete a step in the reasoning process. This mode of explanation is dynamic and is used when the system functions in backward chaining. If the user doesn't understand the reason behind the question. The system is capable of explaining why the question was asked. For example S: Is the following true?

A is the father of B.

U: Why?

S: I need the fact:

A is the father of B. To establish the following fact "B is the son of A"
By using the rule no. 4.

A is the father of B.

B is the son of A.

3. Working memory : It is a global database of facts used by the rules.

4. Knowledge engineering : The primary people involved in building an expert system are the knowledge engineer, the domain expert and the end user. Once the knowledge engineer has obtained a general overview of the problem domain and gone through several problem solving sessions with the domain expert, he/she is ready to begin actually designing the system, selecting a way to represent the knowledge, determining the search strategy (backward or forward) and designing the user interface. After making complete designs, the knowledge engineer builds a prototype. The prototype should be able to solve problems in a small area of the domain. Once the prototype has been implemented, the knowledge engineer and domain expert test and refine its knowledge by giving it problems to solve and correcting its disadvantages.

5. Knowledge base : In rule based architecture of an expert system, the knowledge base is the set of production rules. The expertise concerning the problem area is represented by productions. In rule based architecture, the condition actions pairs are represented as rules, with the premises of the rules (if part) corresponding to the condition and the conclusion (then part) corresponding to the action. Case-specific data are kept in the working memory. The core part of an expert system is the knowledge base and for this reason an expert system is also called a knowledge based system. Expert system knowledge is usually structured in the form of a tree that consists of a root frame and a number of sub frames. A simple knowledge base can have only one frame, i.e. the root frame whereas a large and complex knowledge base may be structured on the basis of multiple frames.

Inference engine : The inference engine accepts user input queries and responses to questions through the I/O interface. It uses the dynamic information together with the static knowledge stored in the knowledge base. The knowledge in the knowledge base is used to derive conclusions about the current case as presented by the user's input. Inference engine is the module which finds an answer from the knowledge base. It applies the knowledge to find the solution of the problem. In general, inference engine makes inferences by deciding which rules are satisfied by facts, decides the priorities of the satisfied rules and executes the rule with the highest priority. Generally inferring process is carried out recursively in 3 stages like match, select and execute. During the match stage, the contents of working memory are compared to facts and rules contained in the knowledge base. When proper and consistent matches are found, the corresponding rules are placed in a conflict set.

Example architecture : Blackboard system

A blackboard system is composed of an area of shared memory, referred to as the blackboard, which contains a problem to be solved and a number of different processes, referred to as knowledge sources that can access and modify the blackboard [Nii].

Each knowledge source will post a partial solution whenever doing so can contribute to the overall solution of the problem. These partial solutions cause other knowledge sources to update their portions of the solution on the blackboard until eventually an answer is found.

Blackboard systems are useful for tasks where different types of knowledge need to be brought to bear: e.g., natural language processing, speech processing, and perception.

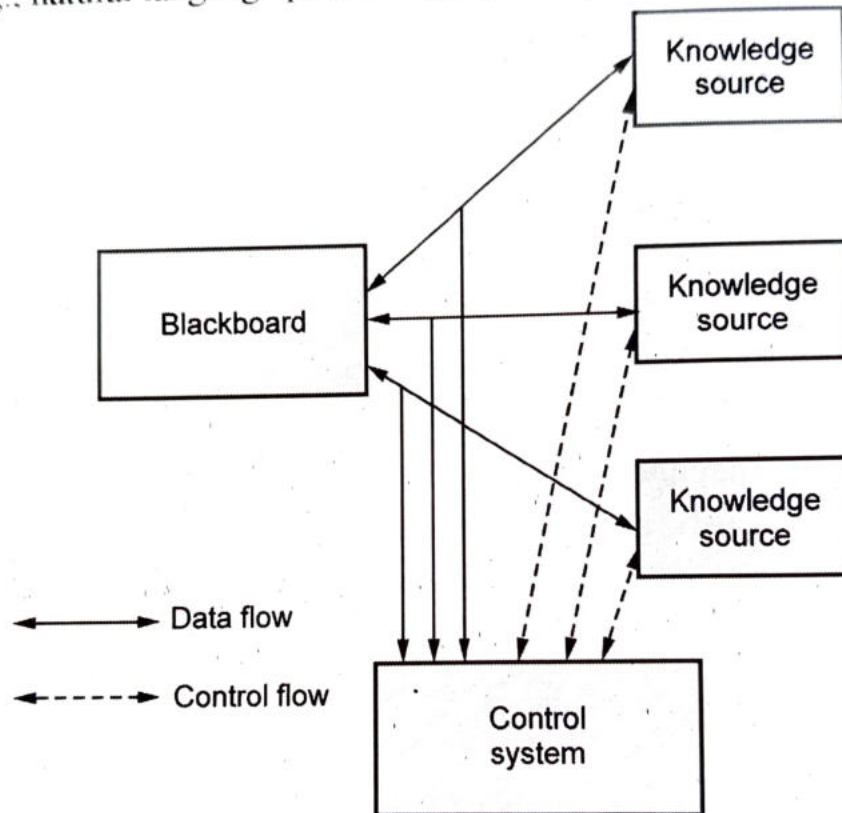


Fig. 2.4.3 Blackboard system

Blackboard system application consists of three major components

1. The software specialist modules, which are called knowledge sources.
2. The blackboard, a shared repository of problems, partial solutions, suggestions, and contributed information.
3. The control shell, which controls the flow of problem-solving activity in the system.

Knowledge sources have a condition-action structure. Ordinarily the condition is a particular configuration of elements on the blackboard. The action ordinarily entails the creation or modification of solution elements on the blackboard.

The advantages of a blackboard include separation of knowledge into independent modules with each module being free to use the appropriate technology to arrive at the best solution with the most efficiency.

Famous examples are the Hearsay-II speech recognition system based on the BBI architecture and Douglas Hofstadter's Copycat program. Other systems include HASP/SIAP that maintained surveillance of surface ships and submarines from sonar data; CRYSLIS, which was designed to infer the three-dimensional structure of protein molecules; and TRICERO, which monitored an area of airspace for traffic. The Intelligent Intermediary for Information Retrieval (IIR), developed by [Croft and Thompson], consists of a group of experts that communicate via a common data structure.

Two free blackboard systems are GBB (UMass blackboard system) and GEST (Generic Expert System Tool). BBTech Corporation sells blackboard systems, multi-agent organizations, and collaborating-software technology solutions.